# Stubbornness and Social Power in Opinion Dynamics: the French–Harary–DeGroot and Friedkin–Johnsen Models

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**Abstract**

~~Abstract.~~ A stubborn agent, which never changes its opinion, is the extreme case of two different mechanisms of resistance to social influence: a self-weight in the French–Harary–DeGroot (FDG) model and an attachment to a fixed prejudice in the Friedkin–Johnsen (FJ) model. Following the tutorial by Proskurnikov and Tempo, this report compares the two models on the same small networks and studies how these mechanisms affect convergence, the final opinions and the influence of each agent, measured by the social power and by the influence centrality. A self-weight acts as inertia: it slows an agent down, makes the network aperiodic and earns influence gradually. A prejudice acts as an anchor: a single prejudiced agent reached by everybody makes the FJ model asymptotically stable and takes the whole influence centrality, while, when every agent is prejudiced, different prejudices keep the group divided on any network. Writing the FJ model as an FDG model on an augmented graph, with the prejudices as virtual stubborn agents, shows that the two models differ in what is a state: the opinion of a stubborn agent is a memory inside the state in the FDG model and an external input in the FJ model.

## Contents

**1 Introduction** **2**
1.1 Preliminaries . . . 2
1.1.1 Graphs . . . 2
1.1.2 Matrices . . . 3

**2 The French–Harary–DeGroot model** **4**
2.1 Introduction to the FDG model . . . 4
2.1.1 The FDG model . . . 4
2.1.2 Consensus and social power . . . 5
2.1.3 Disagreement . . . 6
2.1.4 Rosetta stone . . . 6
2.2 Stubbornness and social power in the FDG model . . . 7
2.2.1 Modelling stubbornness: self-weights and stubborn agents . . . 8
2.2.2 Self-weights, stubbornness and convergence . . . 8
2.2.3 Self-weights, stubbornness and social power . . . 9

**3 The Friedkin–Johnsen model** **11**
3.1 Introduction to the FJ model . . . 11
3.1.1 The FJ model . . . 11
3.1.2 Convergence and stability . . . 11
3.1.3 How the opinions converge: the matrix $V$ and the augmented graph . . . 14
3.1.4 Influence centrality . . . 16
3.2 Stubbornness and influence centrality in the FJ model . . . 16
3.2.1 Modelling stubbornness: prejudices and stubborn agents . . . 16
3.2.2 Prejudices, stubbornness and convergence . . . 20
3.2.3 Prejudices, stubbornness and influence centrality . . . 25

**4 Conclusions** **26**